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Re: Request for Information on the Development of an Artificial Intelligence Action Plan

Introduction

Vantage Data Centers (hereinafter Vantage) appreciates the leadership of the Trump Administration in offering this opportunity for interested parties to provide input through the Request for Information on the Development of an Artificial Intelligence Action Plan. Vantage believes that it is important for the data center industry to be an active participant in discussions about the public policies impacting digital infrastructure and to consider ways in which the United States of America can maintain its leadership role in encouraging economic growth and supporting innovation, and how to win the global competition for artificial intelligence (AI) dominance to support the cause of advancing freedom and prosperity. Vantage is pleased to provide these comments to help inform current discussions within the Administration that will shape the future of digital infrastructure policy and create great opportunities for Americans to enjoy the economic and societal benefits offered by cloud computing and AI.[1]

About Vantage

Vantage serves the leading American technology companies by providing innovative and scalable data center campuses to hyperscalers, cloud providers, and large enterprises. Founded in 2010, we are a privately held, Colorado-based company with approximately 1,800 employees. With 35 operating or developing data center campuses across five continents, Vantage is known for having a strong, collaborative approach in partnering with our customers to develop flexible data center solutions for cloud computing and AI, with the best-in-class customer service. Vantage is expanding within the United States to meet the incredible demand for digital infrastructure, with new hyperscale campuses under development in Georgia, Ohio, Nevada, and Texas, among other locations. Vantage is currently working with several customers on at-scale AI deployments.

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AI Infrastructure – what is needed to keep US at the forefront?

"It is the policy of the United States to sustain and enhance America's global dominance in order to promote human flourishing, economic competitiveness, and national security."[2] President Trump has said on several occasions that with the right public policies, the United States can solidify its position as the world leader in AI development and secure a brighter future for all Americans. **Vantage supports a multi-pronged public policy approach to maintain and secure U.S. leadership on AI, including efforts that encourage additional investment into digital infrastructure, regulatory reforms to unleash more power capacity, promoting workforce upskilling and reskilling, and fostering public-private collaboration.** Vantage recommends the following:

1. Enhance America's power infrastructure to encourage domestic AI data center growth

Enhancing America's power infrastructure is critical to maintain and accelerate U.S. leadership in AI, technological innovation, and the global economy. Meeting the power demand for AI and data center end customers' requirements for reliable, affordable, and innovative power supplies, and ensuring that the next wave of AI innovation is U.S.-driven, requires immediate strategic investments to modernize transmission infrastructure and to build more domestic power generation capacity for data centers.

Currently, transmission capacity limitations pose significant barriers to the growth of digital infrastructure across the U.S. Building an electrical transmission line can take more than a decade due to regulatory and litigation hurdles. For example, in Northern California and Central Washington, some utilities have indicated to data center operators that additional power is at least 7-10 years away. Although data centers have achieved tremendous gains in efficient operations and innovation in electricity delivery, their substantial demand for electrical power remains a primary factor in investment and siting decisions.

Efforts to enhance power infrastructure should specifically target improvements in the planning, permitting, and construction processes for long-range, inter-regional high-voltage transmission lines in the United States. These transmission lines play a critical role in connecting the reliable and timely electricity supplies that data center customers require to high-demand regions, thereby enhancing system reliability, and addressing growing electricity demands from data centers and AI deployments. Policymakers and regulators should prioritize reforms that streamline and accelerate approvals for inter-regional transmission infrastructure, significantly reducing current timelines and better

[2] President Donald J. Trump, Executive Order 14179, January 23, 2025

aligning the nation's grid infrastructure with its long-term economic and technological objectives.

Addressing these constraints also requires significant changes to existing utility business models, which currently lack adequate incentives for rapid and proactive infrastructure expansion. Current regulatory frameworks and traditional cost-recovery models often disincentivize utilities from investing proactively in new generation and transmission capacity. Increasingly, utilities are concerned about risk exposure when expanding generation and grid capacity to meet the needs of data centers. In response, some utilities have begun requiring significant upfront financial assurances from these customers, tying up considerable capital and credit for extended periods. Such requirements chill investment, strand valuable capital, and impede the development of additional data centers and investment in new growing U.S. markets. To address this challenge, the **Trump Administration and Congress should explore federal-level policy interventions—such as loan guarantees, credit insurance, or revolving fund programs—to help utilities manage risk without overly burdening data center operators.** Successful models already exist, including programs administered by the US Department of Energy's Loan Programs Office and the US Department of Agriculture's Rural Utilities Service, which could be adapted or expanded to ease upfront capital burdens, enhance investor confidence, and enable critical infrastructure deployment.

The Trump Administration, along with other U.S. federal, state, and local policymakers and regulators, should also explore and implement reforms that align utility incentives with the strategic need to meet future electricity demand, enhance system reliability, and promote innovation. Areas of focus for these reforms should include encouraging: 1) better transmission planning by utilities and regional transmission organizations; 2) faster processing of interconnection requests by regulators; 3) collaboration among data centers, utilities, and independent power producers; and 4) timely review and approvals of permits.

Substantial investments are urgently needed, not only in transmission infrastructure but also in expanding domestic power generation capacity. The need is so great that data center operators like Vantage have had to take matters into their own hands; recently, we began the operation of a co-located utility-scale, natural gas-powered generation plant in Virginia to power our second Sterling-area campus to meet customer demand. However, this alone is insufficient to address the growing need for more capacity for cloud computing and AI deployments and to meet customer requirements around data center delivery, design, efficiency, and operations. **Achieving this growth depends on comprehensive policy reforms that the Trump Administration should undertake,**

including streamlining permitting processes, enabling and prioritizing behind-the-meter self-generation and co-location solutions, identifying and opening suitable federal lands in proximity to high-demand markets for data center and energy infrastructure development, and rethinking the utility regulatory environment to accelerate the deployment of generation and transmission projects.

The slow pace of infrastructure expansion stems from outdated and prolonged permitting processes, complex regulatory frameworks, and insufficient long-term forecasting of power needs. Comprehensive long-term forecasts of power demand, combined with enhanced transparency, scenario planning, and closer stakeholder collaboration, are essential to addressing these bottlenecks. Utilities must be empowered and incentivized to strategically invest in infrastructure aligned with national economic and technological priorities. Moreover, **the Trump Administration should explore implementing single-window authorities, shot clocks, and preemption to facilitate power infrastructure development to support AI infrastructure deployment**. Such bold permitting reforms have facilitated faster approvals for other essential commercial developments, such as wireless/5G infrastructure, and can readily be applied to the power sector.

Ultimately, a robust power infrastructure—encompassing expanded generation and upgraded transmission, supported by reformed utility business models—is foundational for securing America's continued leadership in AI and other strategically important technologies. Policymakers must urgently prioritize comprehensive reforms that support the modernization and expansion of America's grid, safeguarding economic prosperity, technological innovation, and national security.

2. Build and maintain a robust and resilient supply chain for data centers

Supply chains and trade policies play a crucial role in the operation and expansion of data centers and to AI. The importance of these elements can be understood through various lenses, including economic efficiency, technological advancement, and national security.

Firstly, the efficiency of supply chains directly impacts the cost and speed of data center construction and maintenance. Data centers require a vast array of components, from servers and cooling systems to networking equipment and power infrastructure. Efficient supply chains ensure that these components are delivered on time and at a reasonable cost. Delays or disruptions in the supply chain can lead to significant project delays and increased costs, which can be detrimental to the competitiveness of data centers and, it follows, to the timely development of AI infrastructure. This highlights the need for robust, reliable, and predictable supply chain management to mitigate risks and ensure timely project completion.

Trade policy also plays a significant role in the operation of data centers. Public policies that promote free, fair, and certain trade on critical infrastructure components such as construction materials, electrical transmission equipment, generators, switchboards, chillers, switches, and server racks, among other categories, can significantly lower the costs of building and maintaining data centers for AI. This is crucial for limiting additional costs for data centers, especially those that aim to establish AI dominance.

Furthermore, trade policies that promote the diversification of sourcing options can enhance the resilience of data center supply chains. By increasing domestic manufacturing capacity and diversifying sourcing options, data centers can reduce their reliance on a single source of components and mitigate the risks associated with supply chain disruptions. Additionally, more competition within the industry can promote more product innovation and drive down costs. **Through effective trade policies, the Trump Administration can create a more robust manufacturing base that provides more resilient sourcing options for the data center industry.**

Moreover, trade policies that prevent bad actors from obtaining critical technologies and promote the development of domestic critical mineral resources and semiconductor manufacturing critical to data centers and AI can also enhance the security and resilience of data center supply chains, in furtherance of U.S. AI dominance. **Through thoughtfully developed and implemented export controls that target specific high-risk uses and users, the Trump Administration can effectively promote U.S. technological leadership while limiting access for bad actors.** By investing in and removing barriers to domestic critical mineral development and semiconductor manufacturing, the U.S. can also reduce its reliance on foreign sources of essential components and enhance the security of its data center supply chains.

3. <u>Workforce development to foster faster and better data center development</u>

Data center workforce development in the U.S. is essential for ensuring the efficient and secure operation of data centers. As data centers become increasingly complex and more integral to AI development and other important U.S. industries, having a skilled workforce is crucial not only for building the infrastructure as demand rises, but also for maintaining the uptime, optimizing performance, and implementing advanced technologies such as liquid-to-liquid cooling. A well-trained workforce can effectively manage the hardware, software, and networking components of data centers, ensuring that they operate smoothly and meet the growing, more timely demands of businesses and consumers.

Trades roles play a vital part in data center construction, maintenance, and operations. Electricians, HVAC technicians, mechanical engineers, and other skilled trades professionals are essential. Investing in a skilled workforce across both trades and technology sectors will not only create high-paying jobs but also ensure that data centers remain resilient, energy-efficient, and capable of supporting AI-driven innovation.

To support this goal, **the Trump Administration should consider implementing targeted training programs that focus on the specific skills required for data center construction and operations**. These programs can include courses on data center development, construction, facilities management, cybersecurity, network administration, and emerging technologies like AI. By partnering with educational institutions, industry organizations, unions, and trade schools, as well as online learning platforms, the Administration can ensure that these training programs are accessible and relevant to the needs of the data center workforce.

Another recommendation is to promote apprenticeships and on-the-job training opportunities within the data center industry. Apprenticeships can provide hands-on experience and mentorship, allowing workers to develop practical skills and gain valuable insights from experienced professionals. **The Trump Administration can support this by offering incentives such as tax credits or loans to companies that participate in apprenticeship programs**. This approach not only helps to build a skilled workforce but also fosters a culture of continuous learning and professional development within the industry.

4. <u>Increase network to reduce bottlenecks and develop data centers in rural areas</u>

AI workloads must be connected to the global Internet to operate and have the bandwidth to scale, and there continues to be vast swaths of the U.S. without adequate capacity and density of backbone fiber optic networking.

Fiber optic networks and bandwidth are crucial for data centers, particularly AI data centers, due to their ability to handle vast amounts of data with high speed and efficiency. AI applications require rapid processing and transmission of large datasets, and fiber optics provide the necessary infrastructure to support these demands. The higher speed of fiber optic networks is essential for AI data centers, where real-time data processing and analysis are critical for tasks such as machine learning and predictive analytics.

Moreover, fiber optic networks offer greater bandwidth, which is vital for AI data centers that need to manage and transfer massive volumes of data. High bandwidth ensures that data can be transmitted without bottlenecks, allowing AI algorithms to access and process information seamlessly. This capability is particularly important for applications that involve big data, such as natural language processing and image recognition, where large datasets must be quickly and efficiently processed to generate accurate results.

Fiber optic networks also contribute to the scalability of AI data centers. As the demand for AI applications grows, data centers must be able to expand their network infrastructure to accommodate increasing data loads. Vantage is already responding to customer requests and requirements for 1GW+ campuses with “capacity to expand” in multiple locations in the U.S. Fiber optics provide the flexibility to scale up bandwidth and network capacity without significant upgrades or disruptions. This scalability allows AI data centers to adapt to evolving technological needs and continue to support advanced AI research and development.

Currently backbone fiber networks are concentrated in and around major metro areas. The U.S. must begin to encourage and support the development of more, denser networks into extra-urban and suburban areas and to bring networks to larger, stranded power sources and available land which could be tapped for AI data center development. **The Trump Administration should consider policy mechanisms to facilitate more fiber optic network development, including federal incentives that can incorporated into urban growth plans at state and local levels**. Providing financial incentives such as grants, low-interest loans, and tax credits to network operators and telecommunications companies can significantly reduce the financial burden associated with deploying fiber optic infrastructure in less densely populated regions. These incentives would make it more attractive for such companies to invest in rural areas, ensuring that data centers and rural populations have access to the global Internet in the build out of AI infrastructure.

<u>Conclusion</u>

In conclusion, the development of AI infrastructure is crucial for maintaining the United States' leadership in technological innovation and economic competitiveness. By investing in power infrastructure, ensuring robust supply chains, promoting workforce development, and enhancing network connectivity, the Trump Administration can create a strong foundation for AI advancements in the U.S. and enable U.S. dominance in AI globally. These efforts will not only support the growth of AI technology but also drive growth and innovation, thereby enabling freedom and prosperity. Vantage Data Centers is committed to collaborating with the Trump Administration and other stakeholders to achieve these goals and contribute to the successful future of digital infrastructure and AI leadership by the United States of America.

Sincerely,

John Stephenson
Vice President, Public Policy, Global
Vantage Data Centers